Dishant Modh

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Summary

Site Reliability Engineer and Database Administrator with **4+ years** of experience owning database infrastructure across distributed, cloud-native systems at healthcare-tech and enterprise scale. Core expertise in **PostgreSQL, Aurora, MySQL, MongoDB, Cassandra, and Snowflake** — from schema design and query optimization to multi-region disaster recovery and zero-downtime migrations. Proven track record reducing P1 incident response time by 30%, improving read-heavy query performance by 50%, and driving **$1.2M+ in annualized AWS cost savings** through RDS and EC2 right-sizing. Comfortable owning on-call rotations, post-mortems, and infrastructure-as-code end-to-end via Terraform and GitHub Actions.

Experience

Software Engineer — Site Reliability Engineering (DBA) Bengaluru | Feb 2024 – Present
GlobalLogic (Client: Hitachi) · Enterprise IT & Digital Transformation · Team of 6 SREs

- Owned reliability and performance of **PostgreSQL, Aurora, MySQL, MongoDB, Cassandra, and Snowflake** clusters serving **global enterprise workloads** across 3 AWS regions; maintained 99.95%+ measured uptime against contractual SLA.
- Designed and delivered a **multi-region DR architecture** using Aurora Global Database and AWS DMS, reducing RPO from 15 minutes to near-zero and RTO from 8 minutes to under 60 seconds — validated quarterly through automated failover runbooks.
- Automated RDS provisioning, snapshot scheduling, and patch management via **Terraform + GitHub Actions**, cutting average deployment lead time from ~5 hours to **3 hours** (40% reduction) and eliminating manual config drift.
- Identified and remediated EC2 overprovisioning, idle RDS instances, and unattached EBS volumes through a structured **right-sizing initiative**, delivering **$1.2M in annualized AWS cost savings** (previously reported as $100K/month).
- Built **AWS Lambda + CloudWatch** automation for nightly failover validation and log retention cleanup, reclaiming **10+ engineer-hours per week** previously spent on manual verification.
- Deployed **Datadog + Grafana** centralized observability stack with anomaly-based alerting across all database tiers; reduced Mean Time to Resolution (MTTR) from ~45 minutes to **31 minutes** (30% improvement).
- Optimized **MongoDB aggregation pipelines** (reordering $match/$lookup stages, adding compound indexes) and re-modeled Cassandra partition keys to eliminate hotspots, cutting average query response time from 820ms to **450ms (45% improvement)**.
- Mentored 2 junior SREs on Terraform best practices, on-call runbook authoring, and PostgreSQL vacuum tuning; reduced their mean time-to-first-resolution on database incidents by 35%.

Software Engineer — Database Administrator Ahmedabad | Dec 2021 – Feb 2024
HHAeXchange · Healthcare-Tech SaaS · 500K+ caregivers on platform

- Administered **120+ production PostgreSQL, MySQL, Cassandra, and Oracle databases** across AWS and on-premises environments, processing **2M+ daily transactions** with 99.99% measured uptime.
- Led a deep **PostgreSQL query optimization initiative**: rewrote 40+ slow queries using `EXPLAIN ANALYZE`, corrected missing/bloated indexes, tuned autovacuum thresholds, and eliminated sequential scans — improving read-heavy workload performance by **50%** (p95 latency: 1.8s → 0.9s) and write throughput by **30%**.
- Built Python and PowerShell automation for **streaming replication health checks, table partitioning, and scheduled maintenance** (VACUUM, ANALYZE, REINDEX), reducing DBA manual toil by 40% and eliminating 3 recurring P2 incidents per month.
- Executed **6 zero-downtime on-premises-to-AWS migrations** (PostgreSQL and MySQL to RDS/Aurora) using AWS DMS and pglogical, migrating **4+ TB of production data** with no SLA breach and sub-1-hour cutover windows.
- Consolidated 18 redundant AWS DMS replication tasks and optimized CDC batch sizes and LOB handling, **reducing migration infrastructure costs by 25%** and improving replication bandwidth utilization by 20%.
- Developed reusable **Terraform modules for RDS provisioning** and enhanced Jenkins + Octopus CI/CD pipelines with automated schema migration validation, reducing schema deployment failures to zero over 14 months.
- Served as **primary on-call DBA** for P1/P0 database incidents; authored 12 post-mortems and drove 8 systemic fixes that eliminated repeat incidents, improving team on-call burden by 25%.

Projects

High-Availability PostgreSQL Cluster with Patroni and Terraform github.com/dhmodh

- Deployed a **3-node auto-healing PostgreSQL 15 cluster** on AWS EC2 using Patroni for leader election and etcd for distributed consensus; infrastructure fully managed via Terraform with automated drift detection.
- Integrated **Prometheus + Grafana** dashboards for replication lag, connection pool saturation, and WAL metrics; achieved **< 5-second automatic failover** validated across 20+ simulated failure scenarios.

ML-Powered Database Anomaly Detection (ML-DBA) github.com/dhmodh

- Built a **Python + Scikit-learn Isolation Forest model** trained on 6 months of PostgreSQL `pg_stat_statements` metrics (QPS, lock waits, I/O, cache hit ratio) to predict anomalies including deadlocks, I/O spikes, and autovacuum storms.
- Achieved **91% precision and 87% recall** on held-out production data; reduced proactive investigation time by **60%** and surfaced 4 previously undetected performance regressions in pre-production environments.

Technical Skills

Databases	PostgreSQL, Aurora, MySQL, MongoDB, Cassandra, Snowflake, Oracle
Cloud	AWS (RDS, EC2, DMS, Lambda, S3, IAM, CloudWatch, VPC), Azure (AZ-900)
IaC & DevOps	Terraform, GitHub Actions, Jenkins, Octopus Deploy, Docker, Ansible
Languages	Python, SQL, Bash, Go, PowerShell
Observability	Datadog, Grafana, Prometheus, CloudWatch, Loki, Alertmanager
SRE Practices	On-call rotation, incident command, post-mortems, SLO/SLA management, capacity planning

Certifications

AWS Solutions Architect – Associate	Amazon Web Services
MongoDB SI Associate	MongoDB
Azure Fundamentals (AZ-900)	Microsoft
GitHub Foundations	GitHub

Education

B.Tech in Information Technology 2018 – 2022
Dharmsinh Desai University, Nadiad CGPA: 8.02
Relevant coursework: Database Management Systems, Distributed Computing, Cloud Architecture, Operating Systems